

News Digest

.NET TECHNOLOGY (01CE0523)

MINI PROJECT REPORT

Submitted by:

Enrollment : 92301703046

Enrollment :923017030645

Enrollment : 92301703068

Student Name : Noopur Manva

Student Name : Thun Thingyan Phoo

Student Name : Sasmita Das

BACHELOR OF TECHNOLOGY

in

Computer Engineering



Marwadi
University
Marwadi Chandarana Group



Marwadi University, Rajkot

November, 2025



Marwadi
University
Marwadi Chandarana Group



.NET Technologies (01CE0523)

Mini Project

Faculty of Technology, Marwadi University

Computer Engineering Department

2025-26

CERTIFICATE

This is to certify that the project report submitted along with the project entitled **News Digest** has been carried out by **Noopur Manvar** (92301703046) under my guidance in partial fulfillment for the degree of Bachelor of Technology in Computer Engineering, 5th Semester of Marwadi University, Rajkot during the academic year 2025-26.

Sign:

Prof. Priyanka Mangi

Internal Guide

Sign:

Dr. Krunal Vaghela

Head of the Department



Marwadi
University
Marwadi Chandarana Group



.NET Technologies (01CE0523)

Mini Project

Faculty of Technology, Marwadi University

Computer Engineering Department

2025-26

CERTIFICATE

This is to certify that the project report submitted along with the project entitled **News Digest** has been carried out by **Thun Thingyan Phoo(92301703045)** under my guidance in partial fulfillment for the degree of Bachelor of Technology in Computer Engineering, 5th Semester of Marwadi University, Rajkot during the academic year 2025-26.

Sign:

Prof. Priyanka Mangi

Internal Guide

Sign:

Dr. Krunal Vaghela

Head of the Department



Marwadi
University
Marwadi Chandarana Group



.NET Technologies (01CE0523)

Mini Project

Faculty of Technology, Marwadi University

Computer Engineering Department

2025-26

CERTIFICATE

This is to certify that the project report submitted along with the project entitled **News Digest** has been carried out by **Sasmita Das (92301703068)** under my guidance in partial fulfillment for the degree of Bachelor of Technology in Computer Engineering, 5th Semester of Marwadi University, Rajkot during the academic year 2025-26.

Sign:

Prof. Priyanka Mangi

Internal Guide

Sign:

Dr. Krunal Vaghela

Head of the Department

Index

Acknowledgments.....	i
Abstract.....	ii
1. Introduction.....	1
2. Technology used and Implementation Strategy.....	2
3. Implementation Snapshot.....	6
4. Conclusion.....	11
References.....	12

Acknowledgments

This mini project would not have been possible without the Guidance and the help of several Individuals who in some ways contributed their valuable time and assistance in the completion of this Internship.

We would like to thank my college **Marwadi University** for giving us this opportunity .My sincere gratitude to our Head of The Department (HOD), **Dr. Krunal Vaghela**, my Internal Guide, **Prof. Priyanka Mangi** for helping and solving our queries whenever required.

Abstract

Modern news consumers face information overload due to the vast amount of digital content available. Personalized recommendation has proven effective in helping users find relevant news and alleviating overload. **News Digest** is a .NET/C# web application that delivers personalized news to each user. It displays unfiltered headlines on the Home page (using an external News API) and, upon login, shows a Dashboard of news tailored to the user's interests. The system uses **ML.NET** to perform *sentiment analysis* (classifying news as positive, negative, or neutral) and *topic classification* (e.g. sports, politics) on news text. As users read and react to articles, their preferences (topics and sentiment biases) are tracked in a SQL Server database. Based on this reading behavior and analysis, the app filters and recommends news items most likely to interest each user. Thus, News Digest combines machine learning (ML.NET) with ASP.NET Core to create an intelligent news delivery platform .

1. Introduction

With the explosion of online media, users are inundated with news articles from countless sources. This can lead to information overload and difficulty in finding the most relevant content. Research shows that **personalized news recommendation** significantly improves the user reading experience by helping readers discover news that matches their interests. In this project, we address this need by building *News Digest*, a personalized news web application using .NET technologies. The application's Home page fetches the latest headlines from a **News API**, providing unfiltered, up-to-the-minute news across categories. After a user logs in, the Dashboard displays news that has been filtered according to the user's past interactions and preferences. To model these preferences, the system applies **sentiment analysis** and **topic classification** to every news item. Positive/negative/neutral sentiment scores and category tags are computed for each article. As users click on or spend time reading certain articles, the app records this behavior to infer their interests and sentiment trends. Using this data, the backend personalization logic ranks and recommends articles likely to appeal to the user.

From a technical standpoint, the project leverages the ASP.NET Core framework with C# for the web interface and backend logic. We use **ML.NET** (Microsoft's machine learning library) to build text-classification models. For example, ML.NET's Text Classification capabilities allow us to train models on labeled datasets to predict categories and sentiment in news text without extensive manual preprocessing. In fact, ML.NET can be embedded in an ASP.NET app to classify text in real time. User data (accounts, reading history) is stored in a connected SQL Server database. The overall goal is to apply modern ML techniques in a familiar .NET web stack to deliver a seamless personalized news experience.

2. Technology used and Implementation Strategy

Technology Used :

Technology / Tool	Purpose in Project	Details
ASP.NET Core MVC (C#)	Web application development	Used to build UI (Home, Login, Signup, Dashboard), routing, controllers, and server-side logic.
ML.NET	Machine Learning (Sentiment & Topic Classification)	Performs sentiment analysis (Positive/Negative/Neutral) and topic detection (Sports, Politics, Tech, etc.) on news articles.
NewsAPI	Fetching live news content	Provides external news data from multiple categories and sources. The home page shows unfiltered news.
SQL Server Database	User & reading history	Stores user accounts, login details, reading history, and preference patterns.
Entity Framework Core	ORM for database operations	Manages database CRUD operations and integrates with ASP.NET Identity for authentication.
ASP.NET Identity	Authentication & Authorization	Handles Login, Signup, password hashing, session management.
C# Backend Services	Business logic layer	Handles News API calls, ML model predictions, and recommendation logic.
HTML, CSS, Bootstrap	Frontend UI/UX	For responsive layout of news cards, navbar, dashboard, and forms.

Visual Studio	Development environment	For writing, testing, debugging .NET code and ML integration.
---------------	-------------------------	---

Implementation Strategy

1. Application Structure

- Developed using **ASP.NET Core MVC** with separate layers:
 - Controllers for request handling
 - Models for data representation
 - Views for UI rendering
- Navigation bar includes: **Home, Login, Signup, Dashboard, Contact Us.**

2. Home Page Implementation

- Calls **News API** to fetch latest articles.
- Displays *unfiltered* news to all users.
- News displayed as cards with:
 - Title
 - Image
 - Description
 - Read More link

3. User Authentication

- Implemented using **ASP.NET Identity**.
- Signup stores user details into SQL databases.
- Login validates credentials and creates a secure session.

4. Machine Learning (ML.NET) Integration

- Used **ML.NET Text Classification** .
- ML models (Built in by .NET Framework):

- **Sentiment Analysis Model**
 - **Topic Classification Model**
- Each fetched article is passed through the ML pipeline for:
 - Sentiment label (Positive/Neutral/Negative)
 - Topic label (Sports, Politics, Business, etc.)
- Predictions stored in memory or database for personalization.

5. Dashboard Personalization Logic

- After login, Dashboard loads all news initially.
- System monitors:
 - Which articles user opens
 - Time spent reading
 - Sentiment bias of viewed articles
 - Most-read categories
- Stores user-article interactions in a database table.

6. Recommendation Engine

- Uses past interactions to infer preference patterns.
- Recommends news based on:
 - Frequently viewed topics
 - Sentiment alignment
 - Recency of articles
- Filters & ranks articles before displaying to user.

7. Database Layer

- **SQL Server** stores:
 - Users
 - Article interactions

- Optional preference scores
- **Entity Framework Core** handles database mapping and LINQ queries.

8. API Consumption

- News API integrated using:
 - HTTPClient
 - JSON deserialization
- Supports fetching by category: Sports, General, Tech, etc.

9. UI/UX Implementation

- Built using **HTML + CSS + Bootstrap**.
- Dashboard has:
 - News cards
 - Preference summary section
 - Optional charts/analytics

10. Deployment & Testing

- Tested using a local IIS server.
- Validated:
 - Login security
 - API fetching reliability
 - ML prediction accuracy
 - Database consistency
- Deployment-ready for Windows hosting.

3. Implementation Snapshot

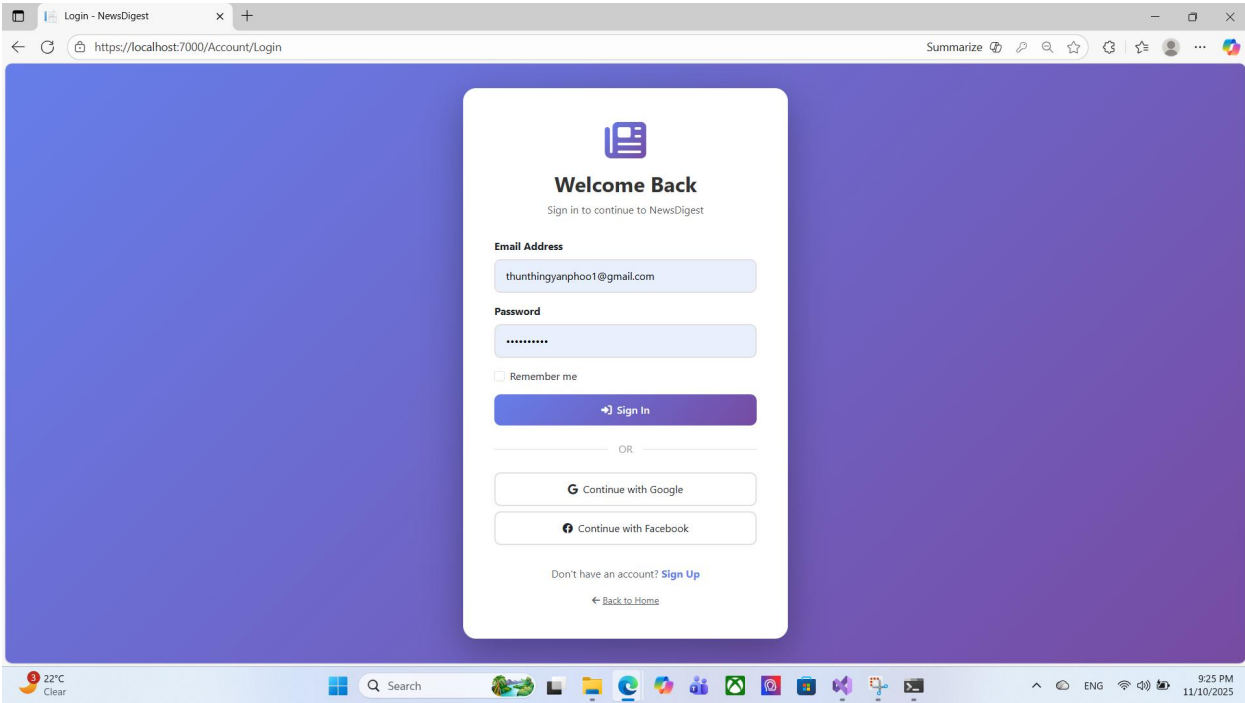


fig1:Login Page

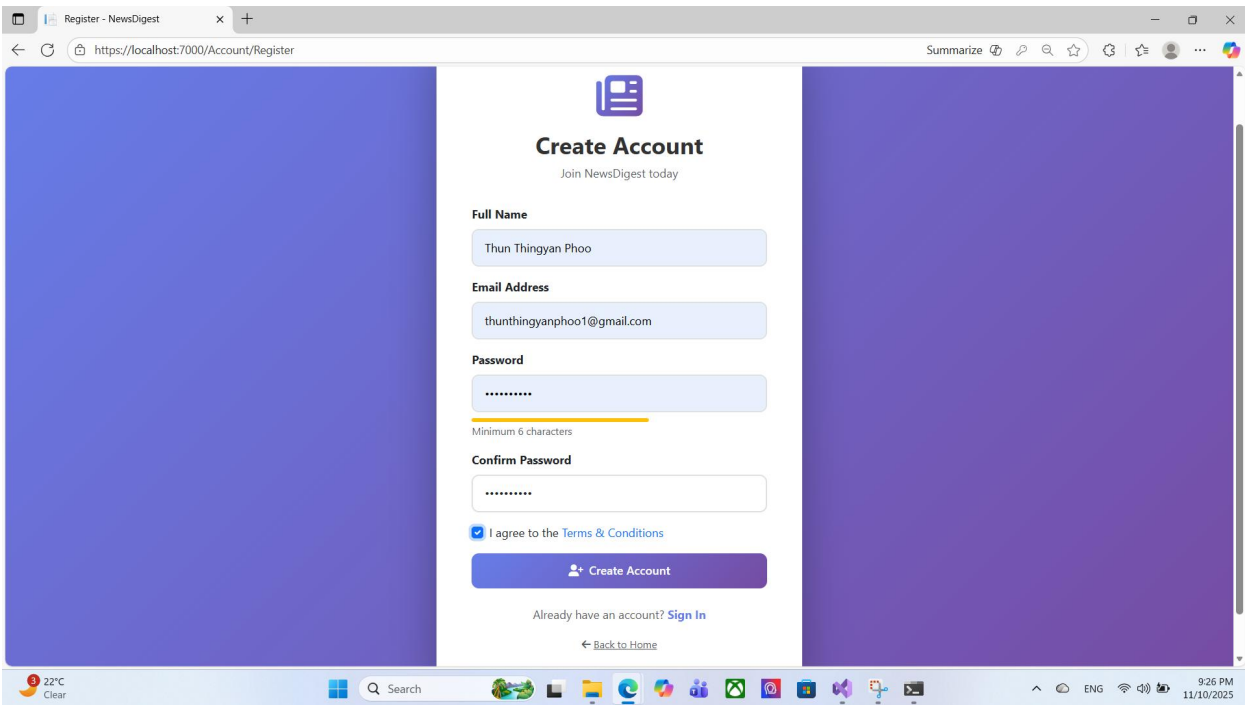


fig2:SignUp page

Home - NewsDigest x +

https://localhost:7000 Summarize

NewsDigest Home Dashboard History Profile Contact

Stay Informed with Smart News

Get personalized news filtered by sentiment analysis powered by ML.NET

Enjoy Scrolling

Latest Headlines

Top technology news with sentiment analysis



Phoronix POSITIVE

The Linux Kernel Looks To "Bite The Bullet" In Enabling Microsoft C Extensions - Phoronix

Two patches queued into the Linux kernel's build system development tree, kbuild-next, would enable the -fms-extensions compiler argument everywhere for allowing GCC and LLVM/Clang to use the Microsoft C Extensions when compiling the Linux kernel

Nov 09, 2025 Read More

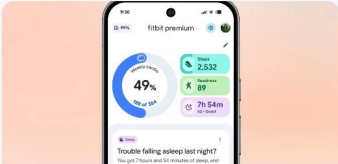


Eurogamer.net NEGATIVE

Players slam Battlefield 6 XP cap: "EA managed to make progression time-gated in a premium game, and nobody's talking about it" - Eurogamer

Battlefield 6 has a daily XP cap and yes, that applies during double XP weekends, too.

Nov 09, 2025 Read More



9to5google.com NEGATIVE

Fitbit redesign brings Material 3 Expressive overhaul - 9to5Google

Besides the AI-powered personal health coach, the Fitbit redesign introduces Material 3 Expressive to the Android app.

Nov 09, 2025 Read More

Why Choose NewsDigest?



AI-Powered

ML.NET sentiment analysis for smart filtering



Personalized

Customize news by interests and sentiment



Real-Time

Get the latest news from NewsAPI

© 2025 NewsDigest - Powered by Marwadi University

Twitter Facebook LinkedIn Instagram

22°C Clear Search 9:25 PM 11/10/2025

fig3:Home Page

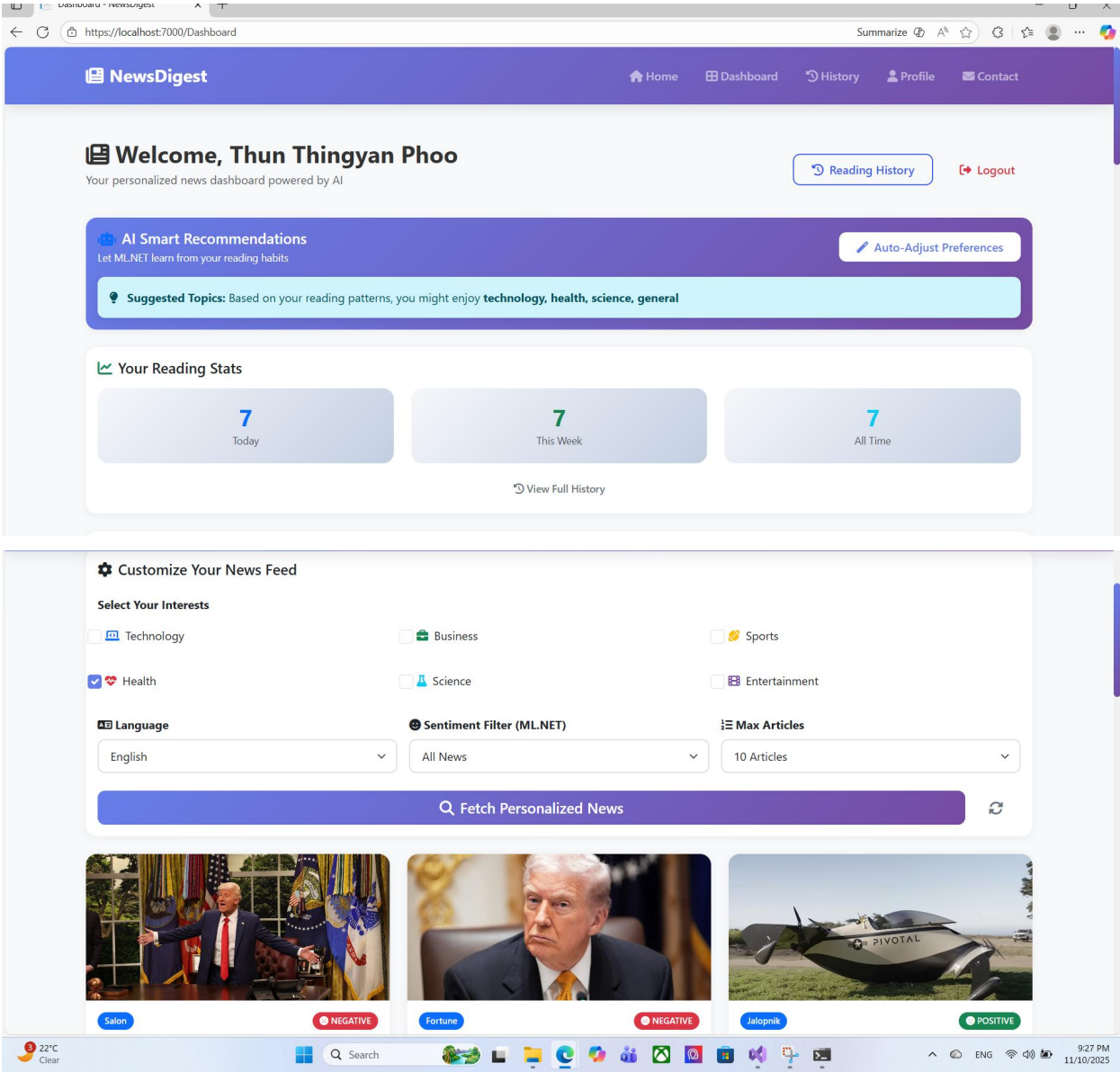


fig4:User's Dashboard page

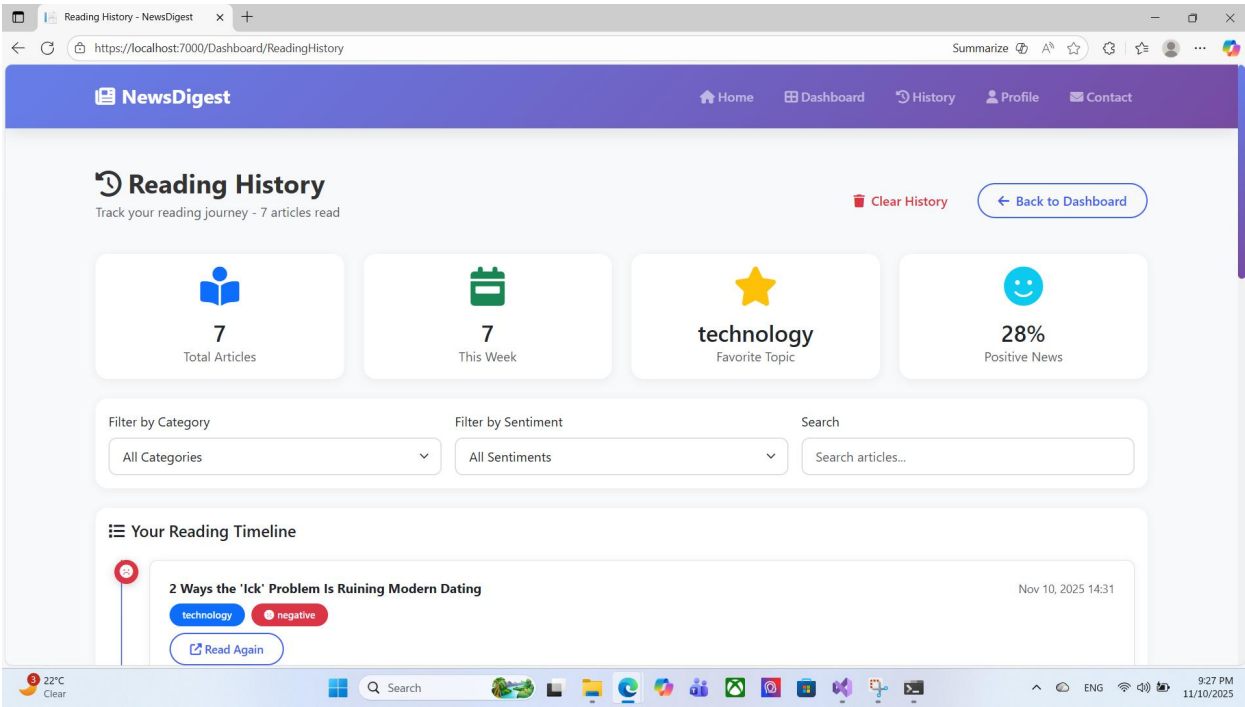


fig5:Reading History Page

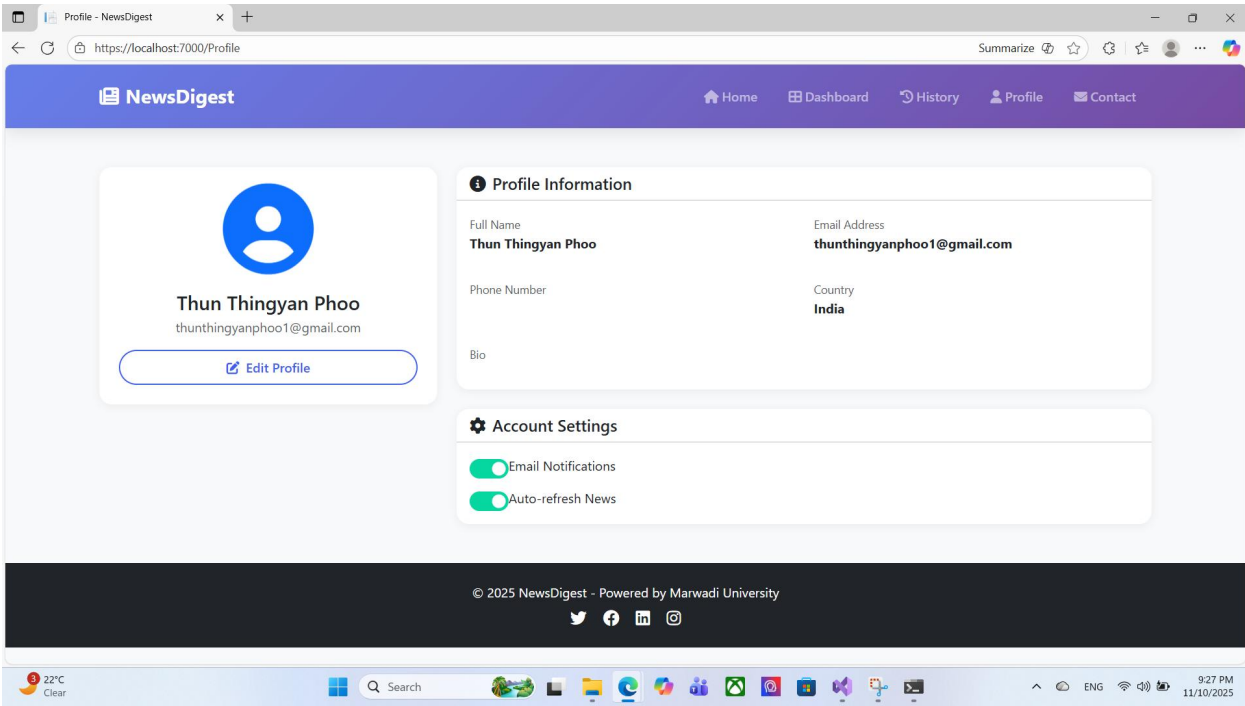


fig6:User Profile Page

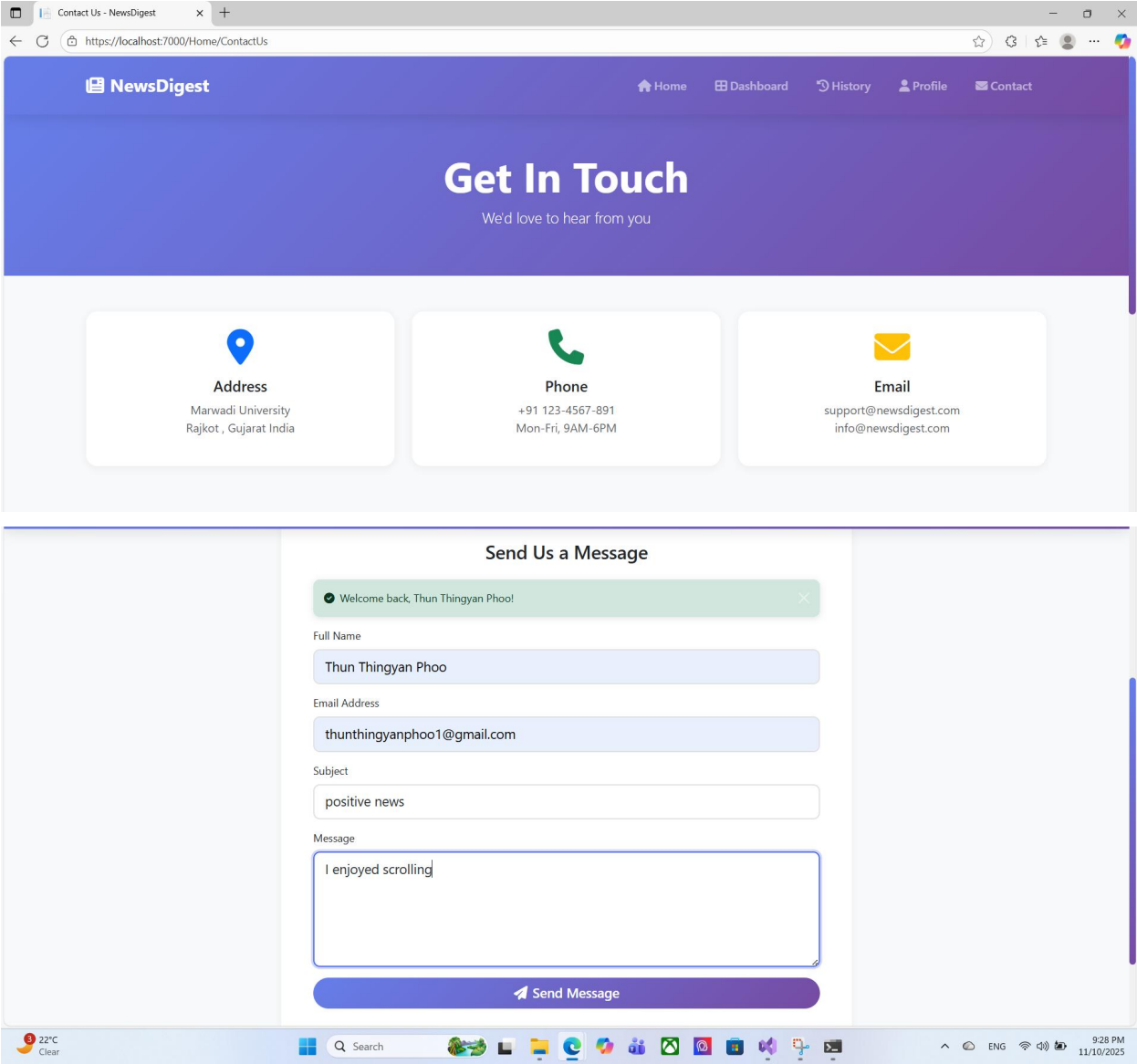


fig7: Contact Us

4. Conclusion

In this project, we developed **News Digest**, a .NET-based web application that demonstrates personalized news recommendation using machine learning. The system successfully integrates a News API for content, ASP.NET Core for user management, and ML.NET for intelligent filtering. Through sentiment and topic analysis of articles, the app learns each user's preferences over time and delivers a custom news feed. This aligns with research findings that personalization can greatly improve user experience in news apps. The modular design also allows for future enhancements: for example, more sophisticated recommendation algorithms (collaborative filtering), real-time updates, or integration of external NLP services. Overall, the project shows how modern .NET technologies can be combined to build a practical, end-to-end application for personalized content delivery.

References

1. Microsoft Learn, *"Tutorial: Analyze sentiment of website comments in a web application using ML.NET Model Builder"*learn.microsoft.com.
2. Matt Eland, *"Text Classification in C# with ML.NET 2.0"* (Accessible AI blog)accessibleai.dev.
3. News API, *"Get Started"* documentation (newsapi.org)newsapi.org.
4. Chuhan Wu et al., *"Personalized News Recommendation: Methods and Challenges,"* *arXiv preprint* (2021)arxiv.org.
5. Microsoft Docs, *"Introduction to Identity on ASP.NET Core"* (Authentication and Identity section)learn.microsoft.com.

